

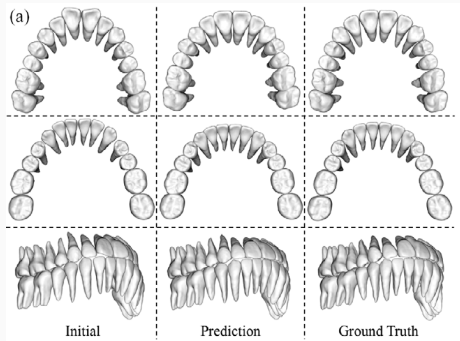
Adapting CLIK-Diffusion to intraoral scans

Tooth alignment prediction

Matteo Princisgh

Artificial Vision

Tooth alignment

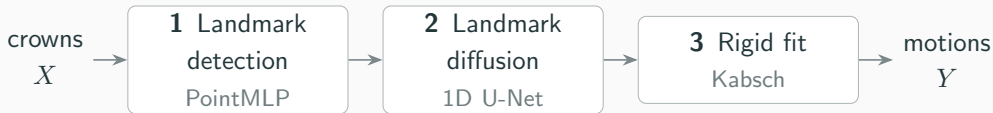


$$h : X \rightarrow Y$$

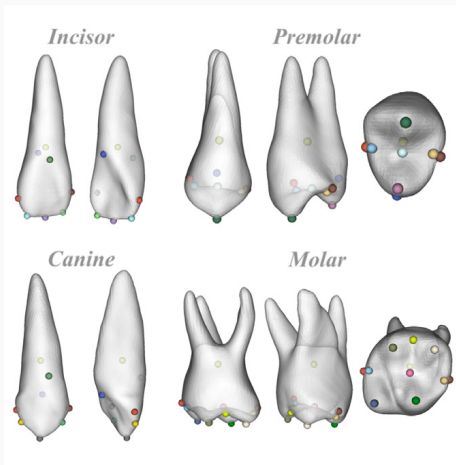
$$X = \{X_i\}_{i=1}^{28}$$

$$Y = \{(R_i, T_i)\}_{i=1}^{28}$$

$$\hat{X}_i = R_i X_i + T_i$$



Stage 1 · Landmark detection

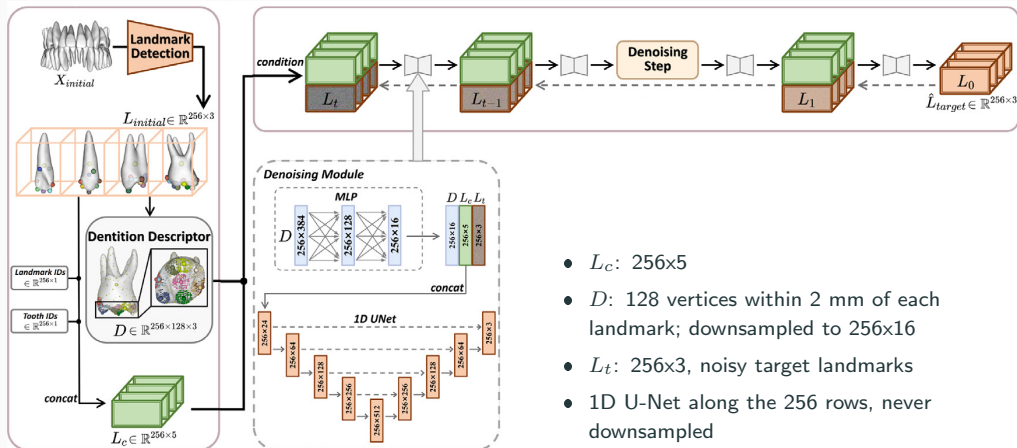


- One PointMLP per tooth class
- 8 / 8 / 9 / 11 landmarks per tooth
→ 256 per dentition

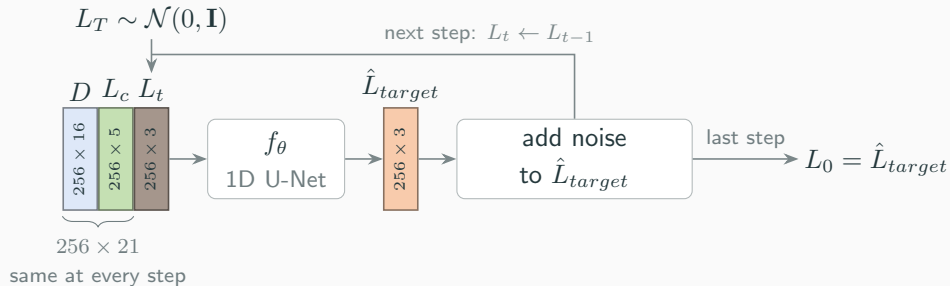
$$L_c = [L_{init} \mid \text{id}_{landmark} \mid \text{id}_{tooth}] \in \mathbb{R}^{256 \times 5}$$

Trained on 1,207 CBCT dentitions annotated by dentists

Stage 2 · Input encoding



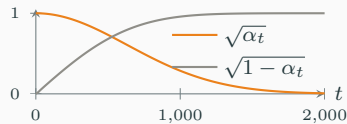
Stage 2 · Diffusion steps



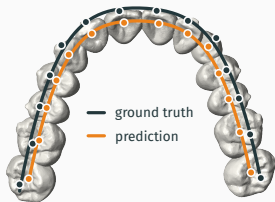
$$q(L_t | L_{t-1}) = \mathcal{N}(\sqrt{\alpha_t} L_{t-1}, (1 - \alpha_t)\mathbf{I})$$

$$q(L_t | L_0) = \mathcal{N}(\sqrt{\alpha_t} L_0, (1 - \alpha_t)\mathbf{I})$$

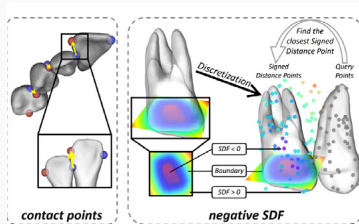
$$T = 2000$$



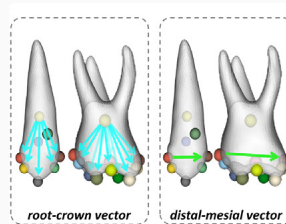
Training · Knowledge-informed constraints



dental-arch level



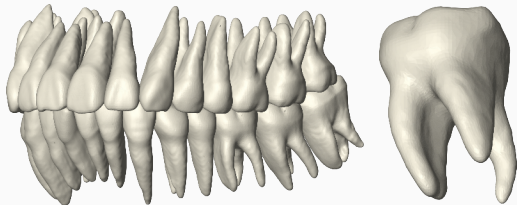
inter-tooth level



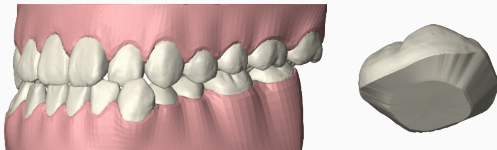
individual-tooth level

Testing on intraoral scans

CBCT



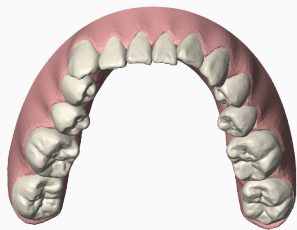
Intraoral scan



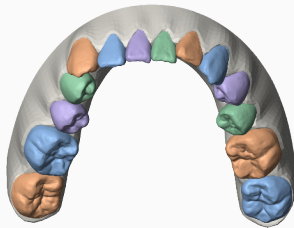
- Wang et al. 2024: 1,060 patients, before and after treatment
- Surface only: gingiva, no roots, crowns closed at the base
- CLIK's authors retrain the landmark detectors on 100 annotated scans

Adapting CLIK to intraoral scans

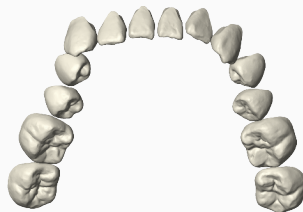
From scans to crowns



arch scan



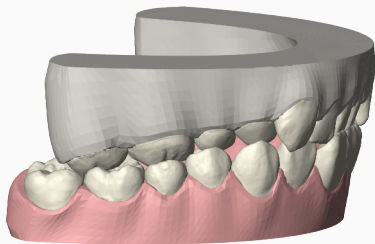
per-vertex tooth labels



one mesh per crown

- FDI numbering → CLIK's 28 tooth slots, third molars dropped
- 158 extraction cases removed → 902 patients

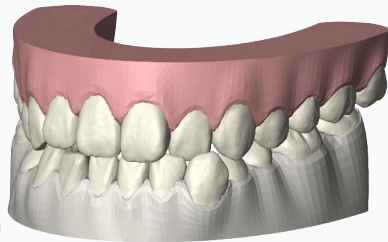
Into CLIK's coordinate frame



as stored



■ upper arch



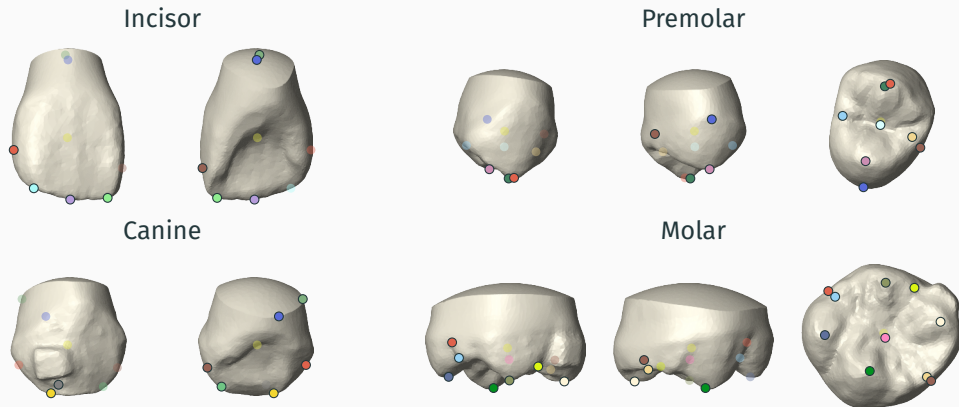
CLIK frame

$$\mathbf{x}' = R(\mathbf{x} - \boldsymbol{\mu})$$

$\boldsymbol{\mu}$: centroid of both arches

$$R = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Landmarks on intraoral scans

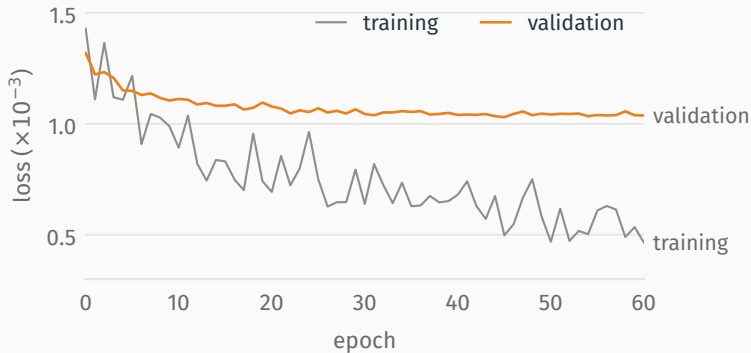


Landmark detector on IOS scans

Finetuning the diffusion model

722 / 90 / 90 patients

$\eta = 10^{-5}$, batch 16



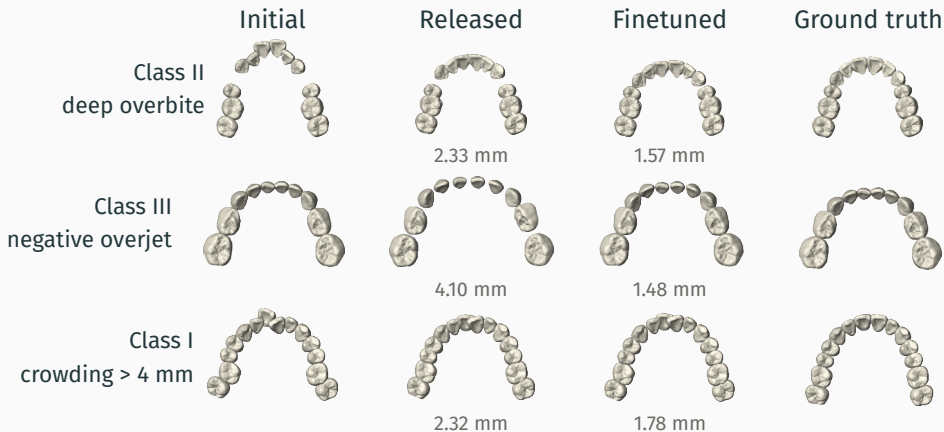
Results

Results on intraoral scans

	Rotation ($^{\circ}$)	Translation (mm)	Point cloud (mm)	Collision (mm)
TANet	11.69 ± 3.19	2.51 ± 0.62	3.59 ± 1.31	0.61 ± 0.21
TANet+	10.88 ± 3.03	2.17 ± 0.55	2.88 ± 1.09	0.70 ± 0.16
TADPM	9.81 ± 3.78	1.77 ± 0.68	1.54 ± 0.72	0.19 ± 0.13
<i>CLIK-Diffusion</i>				
released weights	8.73 ± 2.52	1.75 ± 0.47	1.59 ± 0.71	0.26 ± 0.11
retrained landmarker	6.69 ± 2.56	1.20 ± 0.44	1.30 ± 0.68	0.17 ± 0.13
finetuned diffusion (ours)	7.58 ± 2.71	1.50 ± 0.54	1.30 ± 0.65	0.19 ± 0.11

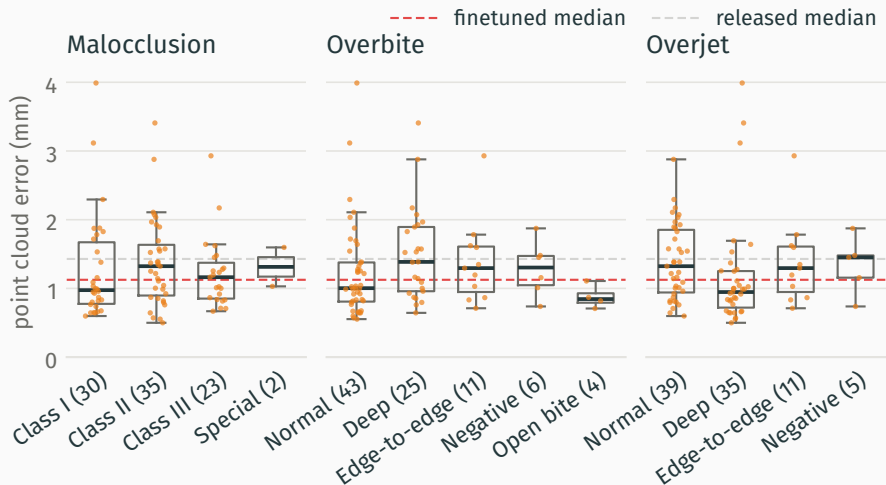
Mean $\pm$ std over the test patients. Other rows: Table 2 of Dou et al.

Clinical cases



One arch per patient; under each prediction, the patient's point cloud error

Clinical cases



- Finetuning only the diffusion model is a viable alternative to annotating scans
- Point cloud error 1.59 $\rightarrow$ 1.30 mm, as low as the retrained landmarker
- No clear difference in performance between clinical cases